

VINAYAK GUPTA

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vinayakgupta29.github.io/portfolio

Professional Summary

Versatile and systems-oriented Software Engineer with 3 years of progressive experience engineering low-latency backend architectures, distributed computing infrastructure, and cross-platform native ecosystems using Go, Node.js (TypeScript), and high-concurrency PostgreSQL. Demanded at the intersection of deep computing and practical software construction, possessing a distinct multi-disciplinary background spanning runtime performance tuning, deterministic discrete mathematical algorithms, structured compiler design, and applied computational linguistics. Proven track record designing resilient cryptographic microservices, hardware-software control loops down to the 8ms execution window, and domain-specific serialization formats that radically eliminate payload network overhead. Adept at navigating legacy API specifications to map multi-layer data tokenization architectures, building multi-threaded database migration engines, and structuring declarative cross-platform mobile/desktop runtimes for applications supporting upwards of 40,000 active users.

Skills & Competencies

Core Systems Languages	Go (Golang), TypeScript, JavaScript, Python, SQL (PL/pgSQL), Java, C++, Rust, OCaml.
Backend & Architecture	Node.js (Express/TS), Django, Echo (Go), RESTful API Design, Microservices Architecture, Event-Driven Automation, Cron Pipelines, Secure Key Management Systems.
Data Layer & Cloud	PostgreSQL (High-Concurrency Optimization, Recursive CTEs, Tree Traversal Models), MongoDB, SQLite, AWS (EC2, S3), Docker, Containerized Environments (Docker Compose), Git Lifecycle.
Client & Hardware Systems	Flutter Engine, Tauri Platform (Native Linux Desktop Application Compilation), React, Embedded Hardware Orchestration (Modbus TCP, Low-Latency PLC Digital Signaling Automation Loops).
Theoretical Computing	Formal EBNF Grammar Design, Lexical Scanners & Hand-Written Recursive-Descent Parsers, Serialization Engine Architecture, Go Reflection API, Data Structure Design.
Mathematics & Analytics	Discrete Mathematical Modeling, Matrix Coordinate Transforms (Pixel-to-Millimeter Spatial Conversions), Sparse-Allocation Bitwise Algorithms, Time-Complexity Optimization ($O(1)$ Scheduling Matrices).
Linguistics & Communication	Natural Language Synthesis, Applied Computational Text Parsing, Multilingual Layout Configuration, Technical System Documentation, Interdisciplinary Cross-Team Project Management, Technical Translation.

Detailed Professional Experience

Techasoft Pvt. Ltd. <i>Systems Engineer (Field Deployment)</i>	Bangalore, India Mar 2026 – Present
- Architected the full-stack system layout and microservice blueprint for industrial manufacturing plant automation, designing a highly resilient Django core backend synchronized with a native, multithreaded Tauri (.deb) frontend compiled to run directly on bare-metal POSIX/Linux client environments. - Engineered high-performance, deterministic hardware-software control loops utilizing Python-based Modbus TCP protocols to orchestrate low-latency 24V Programmable Logic Controller (PLC) signaling networks, achieving a strict, jitter-free sub-8ms execution window to safely drive high-velocity physical conveyor sorting layers. - Formulated a mathematical frame-analysis matrix within Django to compute dynamic pixel-to-millimeter Region of Interest (ROI) spatial coordinate transformations, allowing the vision tracking ingestion pipeline to process real-time structural data streams at continuous 30 frames per second (FPS). - Executed a critical production database migration from an unindexed SQLite file architecture to an enterprise-grade PostgreSQL instance; rewritten complex raw relational queries and established explicit connection pooling parameters to eliminate write-concurrency deadlocks and stabilize data integrity.	
Medoc Health IT <i>Engineering Product Lead & Project Management Intern</i>	Remote Jul 2024 – Jul 2025
Designed and engineered a stateful, zero-external-dependency local Key Management System (KMS) in Go utilizing the Echo microservice framework, implementing custom memory-locked caches and PBKDF2 (Password-Based Key Derivation Function 2) cryptographic master derivations to guarantee key isolation and sub-2ms retrieval latency.	

- Navigated rigorous National Health Authority (NHA) government specifications to map, decrypt, and integrate legacy ABHA (Ayushman Bharat Digital Mission) health data APIs, building a secure translation middleware that processed complex tokenized JSON structural schemas for a secure environment servicing over 40,000 authenticated patients.
- Resolved high-density resource scheduling conflicts by designing a deterministic triage-queuing engine for the MedocEUA emergency application, utilizing an abstract sparse-allocation bitwise slot priority algorithm ($2^n - 1$) to compute instant clinician availability matrices with $O(1)$ temporal complexity.
- Authored cross-platform mobile engine modules in native Flutter, designing low-level background threads to interface with the Android Health Connect API alongside decentralized user-consent cryptosystems that ensure zero-knowledge data synchronization across active devices.

Appable Technologies

Freelance Full-Stack Software Developer

Remote

Jul 2023 – Jun 2024

- Re-architected multiple legacy client applications by executing complete platform-migrations away from high-cost, server-less Google Firebase infrastructure into standalone Node.js (Express/TypeScript) runtimes backed by specialized PostgreSQL clusters, cutting clients' operational infrastructure overhead by up to 65%.
- Solved deep hierarchical querying issues within an enterprise Human Resource Management System (HRMS) by implementing recursive PostgreSQL Common Table Expressions (CTEs) and Modified Preorder Tree Traversal (MPTT) structures to compute nested, infinite-depth supervisor approval permissions instantly.
- Built an offline-first geofenced attendance verification system implementing local camera biometrics, cryptographic geographic validation hashes, and real-time shift-delta analytics pipelines managed by automated, transactional background cron engines to process bulk system adjustments safely.
- Designed a high-traffic sports booking engine featuring dynamic, multi-variable pricing grids (factoring hourly demand, peak modifiers, and membership tiers), integrating custom Google Maps spatial geometry layers and asynchronous, failure-tolerant SMS invitation fallbacks for un-registered application guests.

Medoc Health IT

Software Development Intern

Remote

Aug 2022 – Jul 2023

- Contributed to the development, stability, and patching of core health informatics software using Flutter, Node.js, and raw SQL relational querying, writing modular, testable UI elements and backend endpoints.
- Optimized individual local developer environment setup times by writing multi-stage Dockerfiles and containerized orchestration layers using Docker Compose, establishing clean development-to-production parity.

Projects & Computational Artifacts

Zundrath Key Management System

Technologies: Go, Echo Framework, Cryptography, PBKDF2, Mutex Design, Storage Systems

- Developed a local-first, low-overhead cryptographic microservice written entirely in Go to serve highly concurrent local applications with sub-2ms storage lookup and key-extraction capabilities.
- Implemented rigorous multi-threaded safety protections using standard `sync.RWMutex` memory configurations alongside an on-demand custom disk-indexing scanner to eliminate the need for memory-heavy external database layers.
- Designed a fault-tolerant, two-phase key rotation transaction protocol that supplies a fallback 24-hour safe-migration window, guaranteeing zero decryption blackouts or race conditions during live schema updates.

Agora.Abhivyakti (Stateless Textboard Architecture)

Technologies: Go, HTML/Template Engine, Applied Linguistics, Cryptographic Hashing, Proxy Proxies

- Built a high-performance, completely stateless live-hosted web forum utilizing Go's native `html/template` compilation engine to deliver fully server-side rendered (SSR) structural layouts, eliminating all state storage overhead.
- Engineered an un-spoofable tripcode identification system utilizing custom cryptographic hashing routines that condense user secret passphrases into unique 16-character public identifiers matching restricted, regex-validated alphanumeric strings ($[A-Z][a-z][0-9]$).
- Integrated an intermediate linguistic validation and proxy security layer that intercept and parses all submitted Markdown formatting, executing strict sanitization loops and mandatory outbound-redirect warning wrappers to insulate platform visitors from malicious injection.

Grounded Object Data (GOD) Format & Compilation Suite

Technologies: Go, EBNF Grammar Design, Lexical Scanners, Recursive-Descent Parsing, Reflection API

- Conceived, specified, and engineered a completely custom, schema-less data serialization format and compiler built on a strict, formal Extended Backus-Naur Form (EBNF) grammar specification, specifically optimized to eliminate token redundancy inherent in JSON payloads.
- Hand-wrote a highly optimized lexical analyzer (scanner) and recursive-descent parser in Go, introducing a novel first-class structural paradigm for multi-row data representations: a tabular syntax (`header:rows`) that completely drops repeating object-key declarations from multi-record streams to optimize transport pipelines.

- Implemented absolute "grounded" type-safety semantics inside the runtime unmarshalling engine via Go's native reflection API, structurally preventing null-pointer runtime exceptions by auto-initializing missing, undefined, or corrupt incoming streams to zero-values or explicit \0 null-terminators.

Publications

Book : Poetry Collection

Title : *Life*

Author : Vinayak Gupta

Published by : Sangam Publications, 2023

Patents

Vinayak Gupta (Co-author)

Title : *System for Cooking*

Application ID : 202521070227

Indian Patent Office, Published, 2025

Research Work

Research Paper Draft

- Title : *A Novel approach to Encryption for Data Security in Mobile Applications*
- Status : Manuscript draft under preparation.
- Area : Backend systems / cryptography / Secure Systems design.

Writing & Creative Work

- Maintain technical and general writing through blogs (topics include software, systems, and analytical content).
- Write fiction (short stories and narrative pieces; selected works to be published on AO3).
- Interested in combining storytelling and technical communication for educational and conceptual clarity.

Research Interests

- Early-stage independent exploration of Autism, focusing on both scientific understanding and support frameworks.
- Interest in accessibility, cognitive diversity, and designing systems for neurodivergent users.
- Drafting a research paper (manuscript in progress).

Certifications

Cyber Security and Ethical Hacking

IGTR, Indore — 2025

Education

B.Tech in Computer Science and Engineering

LNCT University, Bhopal

2021 – 2025

Technical Interests

Backend architecture, distributed systems, applied cryptography, accessibility-focused systems, programming language design